

Automated Accessibility Findings Report (Graded)

WCAG 2.1 Level AA – Preliminary Discovery

Project Information (Not Graded)

Course: CYBR-2600 Cyber Security & Network Practicum

Client Organization: Village of Jacksonville, Ohio

Website Reviewed: jacksonvilleoh.com

Student Team Members:

- Cameron McKenzie

Instructor: Norma Depriest

Testing Week: Week 3

Section 1: Purpose of Automated Testing (10 points)

Instructions

Explain the purpose of automated accessibility testing in the context of a professional audit. This section should be written for a **non-technical audience** and should clearly state the **limitations** of automated tools.

Required Elements

- Purpose of automated testing as *initial discovery*
- Statement that automated testing does **not determine full compliance**
- Reference to later manual validation

Student Response

Automated accessibility testing is used as an initial discovery step in a professional accessibility audit. Automated tools can quickly scan web pages and identify common accessibility issues such as missing alternative text, color contrast problems, or structural issues in the page markup. These tools help highlight potential barriers that may affect users with disabilities. However, automated testing alone cannot determine full accessibility compliance. Many accessibility issues require human review, such as evaluating the meaning of alternative text or testing keyboard navigation. Because of this, automated findings are used as a starting point and are later validated through manual testing.

Grading Criteria (10 pts)

- **10 pts:** Purpose and limitations clearly explained in plain language
 - **7 pts:** Purpose explained but limitations unclear
 - **4 pts:** Vague or partially incorrect explanation
 - **0 pts:** Purpose not addressed
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Section 2: Pages Tested (10 points)

Instructions

List the representative pages tested during automated scanning. Do **not** list every page on the site—only pages approved in the audit plan.

Page Inventory

Page Type	URL or Description	Reason for Inclusion
Homepage	https://jacksonvilleoh.com/	The homepage was included because it is the main entry point to the website and represents the overall structure and navigation used throughout the site.

News / Minutes	https://jacksonvilleoh.com/minutes/	This page was included because it contains frequently updated public information and links to documents used by residents.
Department Page	https://jacksonvilleoh.com/category/utilities/	This page was selected as a representative example of a departmental or service-related section of the website.
Contact Page	https://jacksonvilleoh.com/directory/	The directory page was included because it contains important contact information that users rely on to communicate with local government.
PDF Document	https://jacksonvilleoh.com/wp-content/uploads/2026/03/Council-Minutes-02-25-26-Regular.pdf	This document was included because government websites commonly publish information through PDFs, and document accessibility is important for public access.

Grading Criteria (10 pts)

- **10 pts:** Page selection is logical and aligns with audit plan
 - **7 pts:** Page selection mostly appropriate
 - **4 pts:** Page selection unclear or poorly justified
 - **0 pts:** Pages not identified
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Section 3: Tools Used (10 points)

Instructions

Identify the automated tools used and briefly explain **what they were used to detect**.

Required Elements

- Name or description of tools
- Clear explanation of what automated tools detect well
- Avoid naming tools without explanation

Student Response

The primary tool used during this automated accessibility scan was the WAVE Web Accessibility Evaluation Tool developed by WebAIM. WAVE analyzes the structure and content of a web page and identifies potential accessibility issues that may affect users with disabilities. The tool is particularly useful for detecting problems such as missing alternative text for images, color contrast issues, empty links, and structural problems in page markup. Automated tools like WAVE help quickly identify common accessibility barriers, allowing auditors to focus on the most likely problem areas before conducting manual testing.

Grading Criteria (10 pts)

- **10 pts:** Tools and purpose clearly explained
 - **7 pts:** Tools identified with minimal explanation
 - **4 pts:** Tools listed but poorly explained
 - **0 pts:** Tools not identified
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Section 4: Summary of Automated Findings (15 points)

Instructions

Summarize the **types of issues identified**, not just the number of issues. Focus on **patterns**, not counts.

Required Elements

- Description of common issue categories
- Plain-language explanation of why these issues matter
- No legal or compliance conclusions

Student Response

The automated scans identified several recurring accessibility issues across the pages tested. One common issue involved linked images missing alternative text, which can prevent screen reader users from understanding the purpose of those images or links. Another pattern involved color contrast warnings and structural alerts within page navigation and content sections. These issues may make it more difficult for users with visual impairments or assistive technologies to interpret page content or navigate the website effectively.

The scans also identified issues related to empty links and missing or unclear page titles in some locations. While automated tools cannot determine whether every flagged issue represents a true accessibility barrier, the patterns identified suggest areas where further manual testing and accessibility improvements may be needed.

Grading Criteria (15 pts)

- **15 pts:** Clear, professional summary focused on patterns
 - **10 pts:** Summary present but overly technical or list-based
 - **5 pts:** Summary vague or misleading
 - **0 pts:** Summary missing
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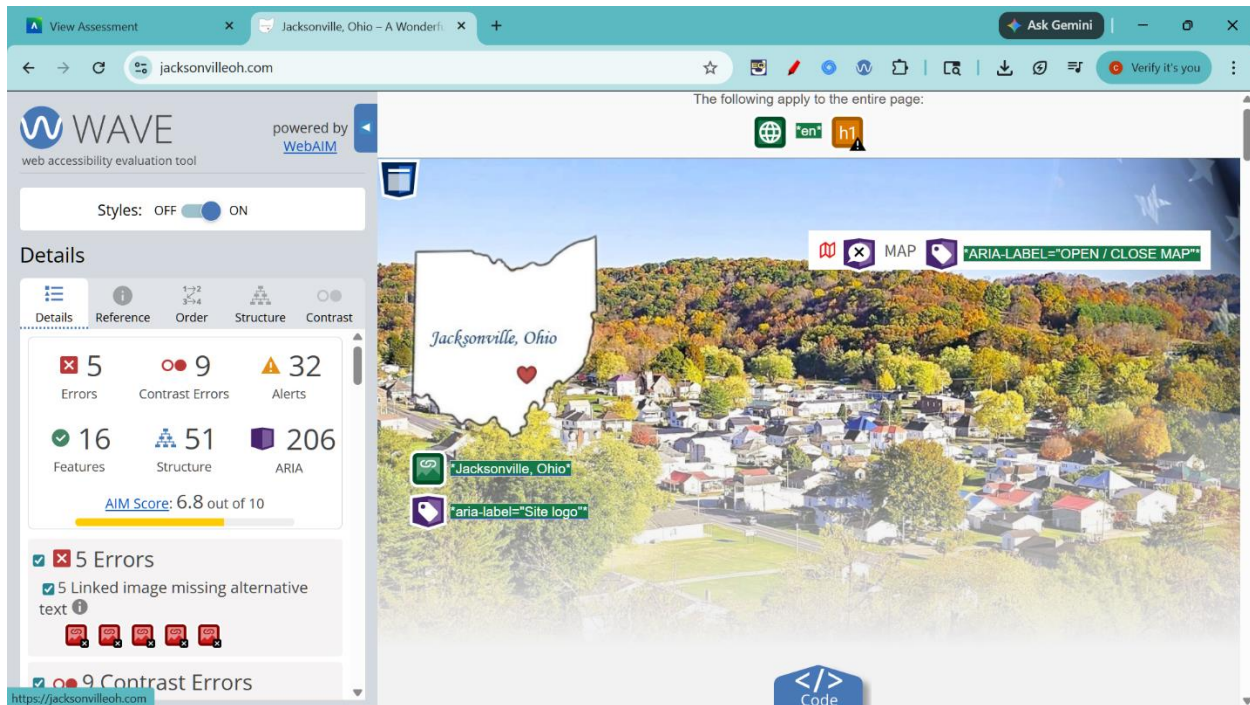
Section 5: Detailed Findings (35 points)

Instructions

Document **individual accessibility issues** identified through automated testing. Be selective—quality matters more than quantity.

Finding #1

Screenshot:



Page:

<https://jacksonvilleoh.com>

Location on Page:

Homepage banner and linked images

Issue Description:

Several images used as links on the homepage are missing alternative text. Alternative text allows screen reader users to understand the purpose or meaning of images on a page.

WCAG Principle (POUR):

Perceivable

Evidence:

Automated testing using the WAVE tool flagged multiple linked images missing alternative text on the homepage.

Tool Used:

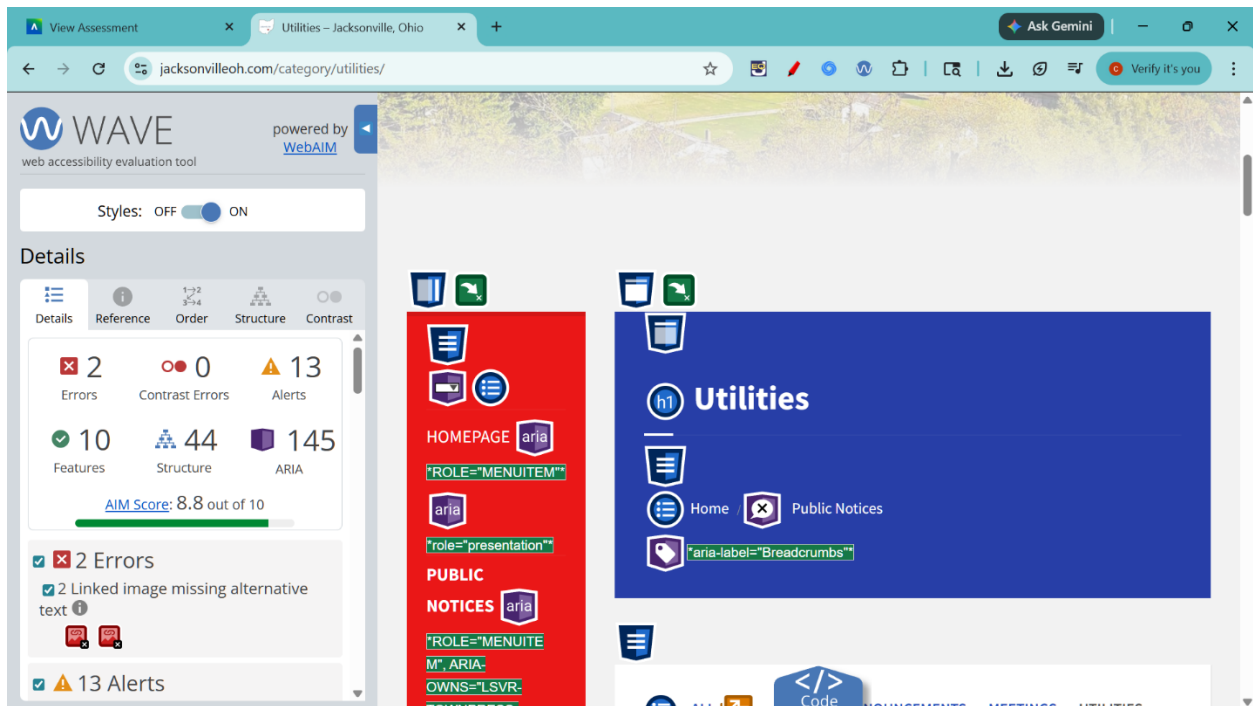
WAVE Web Accessibility Evaluation Tool

Potential User Impact:

Users who rely on screen readers may not know what the linked images represent or where the links lead, which can make navigation confusing.

Finding #2

Screenshot:



Page:

<https://jacksonvilleoh.com/minutes/>

Location on Page:

Meeting Minutes page navigation and page content

Issue Description:

The automated scan identified several color contrast warnings between text and background elements on the page.

WCAG Principle:

Perceivable

Evidence:

The WAVE scan detected multiple contrast warnings on the Meeting Minutes page.

Tool Used:

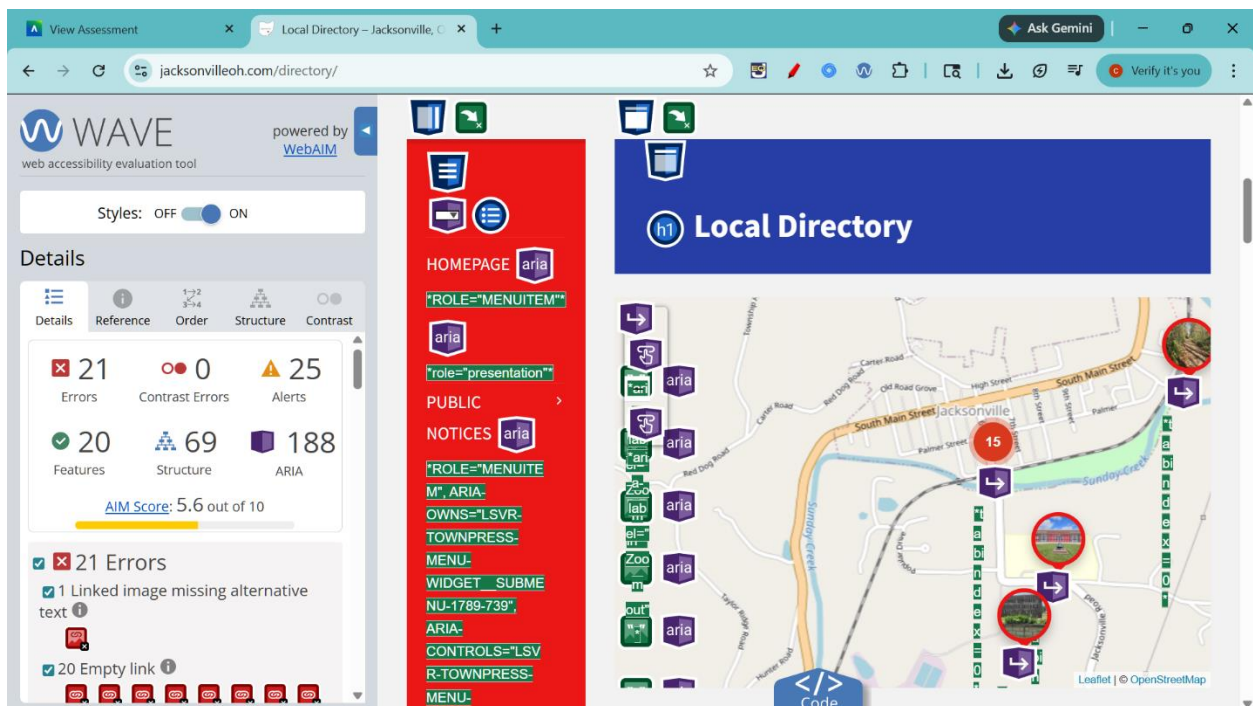
WAVE

Potential User Impact:

Low contrast text can make content difficult to read for users with low vision or color vision deficiencies.

Finding #3

Screenshot:



Page:

<https://jacksonvilleoh.com/directory/>

Location on Page:

Directory page navigation elements and interactive components

Issue Description:

The automated scan detected empty links within the page structure. Empty links occur when clickable elements do not contain meaningful text or labels.

WCAG Principle:

Operable

Evidence:

WAVE flagged multiple empty link errors on the directory page.

Tool Used:

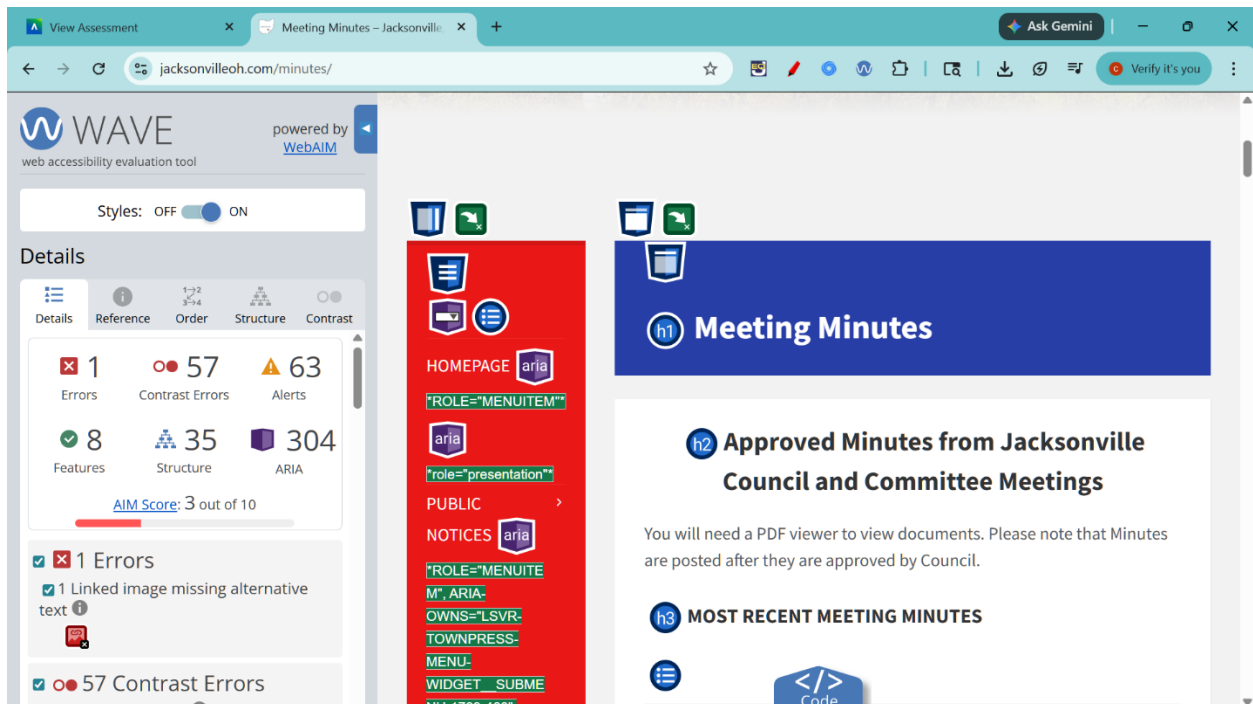
WAVE

Potential User Impact:

Screen reader users may encounter links that do not clearly indicate their purpose, making navigation difficult.

Finding #4

Screenshot:



Page:

<https://jacksonvilleoh.com/category/utilities/>

Location on Page:

Utility category page navigation and linked images

Issue Description:

Some linked images on the utilities page were flagged as missing alternative text.

WCAG Principle:

Perceivable

Evidence:

WAVE detected linked images without alternative text during the automated scan.

Tool Used:

WAVE

Potential User Impact:

Without descriptive alternative text, screen reader users may not understand the purpose of these images or associated links.

Finding #5

Screenshot:

The screenshot displays the WAVE (Web Accessibility Evaluation Tool) interface overlaid on a web browser. The browser window shows the URL `jacksonvilleoh.com/wp-content/uploads/2026/03/Council-Minutes-02-25-26-Regular...`. The WAVE tool's sidebar on the left indicates a "2 Errors" status. The "Details" section lists the following errors:

- 1 Missing or uninformative page title
- 1 Language missing or invalid

The main content area of the browser shows the "Village of Jacksonville Regular Council Meeting February 25, 2026 6:30 PM". The text describes the meeting proceedings, including the presence of Mayor Sam Smathers, Council members, and the approval of minutes. The WAVE tool's "AIM Score" is displayed as 6.7 out of 10.

Page:

<https://jacksonvilleoh.com/wp-content/uploads/2026/03/Council-Minutes-02-25-26-Regular.pdf>

Location on Page:

PDF document structure

Issue Description:

The automated scan indicated that the PDF document may have structural accessibility issues, including a missing or unclear page title and missing language information.

WCAG Principle:

Understandable

Evidence:

The automated tool flagged issues related to page title and language information when the PDF was scanned.

Tool Used:

WAVE

Potential User Impact:

Users relying on screen readers may have difficulty understanding document structure or language settings without proper metadata.

Grading Criteria (35 pts)

- **35 pts:** Findings are clear, well-documented, and evidence-based
 - **25 pts:** Findings mostly clear with minor documentation gaps
 - **15 pts:** Findings poorly explained or weakly supported
 - **0 pts:** Findings missing or unusable
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Section 6: Evidence Quality & Documentation (10 points)

Instructions

Explain how evidence was collected and why it supports the findings.

Required Elements

- Reference to screenshots or captures
- Page context provided
- Clear link between issue and evidence

Student Response

Evidence for each finding was collected using screenshots generated during automated testing with the WAVE accessibility evaluation tool. Each screenshot shows the page being tested along with the accessibility indicators highlighted by the tool. The screenshots also include the page URL and the WAVE interface, providing clear context for where the issue occurs. This documentation allows the findings to be easily reviewed and validated during later manual testing.

Grading Criteria (10 pts)

- **10 pts:** Evidence collection is professional and reproducible
 - **7 pts:** Evidence present but incomplete
 - **4 pts:** Evidence poorly described
 - **0 pts:** Evidence missing
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Section 7: Limitations of Automated Testing (10 points)

Instructions

Describe **why automated testing alone is insufficient** and what will be addressed in manual testing.

Required Elements

- At least two limitations of automated tools
- Reference to keyboard or screen reader testing
- Professional, non-critical tone

Student Response

Automated accessibility testing has several limitations and cannot identify every accessibility issue on a website. Automated tools can detect technical problems such as missing alternative text, empty links, or color contrast issues, but they cannot determine whether content is meaningful or usable for people with disabilities. For example, automated tools cannot evaluate whether alternative text accurately describes an image or whether page navigation is logical for screen reader users.

Because of these limitations, automated testing is only one step in the accessibility auditing process. Additional manual testing, including keyboard navigation testing and screen reader testing, is necessary to fully evaluate the accessibility of the website.

Grading Criteria (10 pts)

- **10 pts:** Limitations clearly and accurately described
 - **7 pts:** Limitations mentioned but not well explained
 - **4 pts:** Limitations vague or incorrect
 - **0 pts:** Limitations not addressed
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 **TOTAL POINTS: 100**

Section	Points
Purpose of Testing	10
Pages Tested	10
Tools Used	10
Findings Summary	15
Detailed Findings	35

Evidence Quality	10
Limitations	10
Total	100

Instructor Review & Approval

☐ Approved ☐ Revisions Required

Instructor Comments:
